ILT Research System - Software Requirements Specification & Functional Design (SRS / FRS v3.3)

Functional Requirements, UI Wireframes, Workflows, Analyst Portal & Replay Engine, Open-Source Cohort Analytics, and Clinical Graphomotor Models

Document Parameter	Specification Value	Document Parameter	Specification Value
Document ID:	ILT-FRS-V3.3-EN	Publication Date:	October 2026 (Release 3.3 Full Edition)
Clinical Basis:	Prof. Adi-Japha, Dr. Hollander, Dr. Dorfberger	Hardware:	Wacom MovinkPad 11 (Continuous Charging OK)
Analyst Capabilities:	Individual Session Replay + Open-Source Cohort Slices	Target Cohort:	Kindergarten to All Elementary Grades
Video Integration:	Agnostic RTSP/ONVIF/USB (Admin Disable Switch)	Timer Governance:	20s System Default (Director Auth Required)

1. Introduction, Objectives, and Clinical Research Foundation

An open-source non-commercial educational research platform for early screening and remediation of learning disabilities across kindergarten and all elementary school grades, with Low-SES focus.

- **Clinical Validation Pilot Cohort:** 50 children on digital tablet, 50 on standard paper, and 50 legacy historical datasets.
- **Continuous Charging Support:** Wacom MovinkPad 11 certified for continuous operation while plugged into USB-C PD power.

2. Summary of Clinical & Operational Decisions (Release 3.3)

#	Domain	Approved Policy	Technical Impact
1	Analyst Portal, Kinematic Replay & Cohort Analytics	Analyst possesses independent database access (de-identified, not present during test). Ability to replay full stroke-by-stroke kinematics at 0.25x-4x speeds synced with desk video, and perform multi-dimensional cohort slice-and-dice using open-source tools (Apache Superset, DuckDB, Jupyter).	Dedicated Analyst Portal web app, Canvas kinematic replayer, analytical database views, and Parquet/DuckDB export pipeline.
2	3 Distinct Clinical Metrics (Prof. Esther Adi-Japha)	System preserves and reports 3 independent metrics: (1) Pen Lifts, (2) Accuracy %, (3) Total Time (w/ corrections), without collapsing into a single Z-score.	Distinct database columns; Comparative Summary Dashboard tracking 3 metrics independently against 4 anchor pairs.
3	Timer Governance & Director	In-Transition = 20s, Inactivity	Session snapshots; Director

	Authorization	Timeout = 20s. Domain overrides require Research Director authorization code and Admin tooltip.	authorization lock; analytical timer factoring enabled in BI queries.
4	Camera-Agnostic Ingestion & Video-Optional	Agnostic RTSP/ONVIF/USB ingestion (desk/hands only). System Admin toggle for video-optional administration.	go2rtc ingestion proxy; video_recording_enabled boolean flag in schema.
5	Unlimited Dynamic Templates Engine	Unlimited custom templates (10-line, 14-line, 1-line, custom grids, blank) across all elementary grades.	Table dynamic_templates with JSON ruling geometry and kinetic movement rules.

3. Clinical Workflows & Research Analyst Subsystem

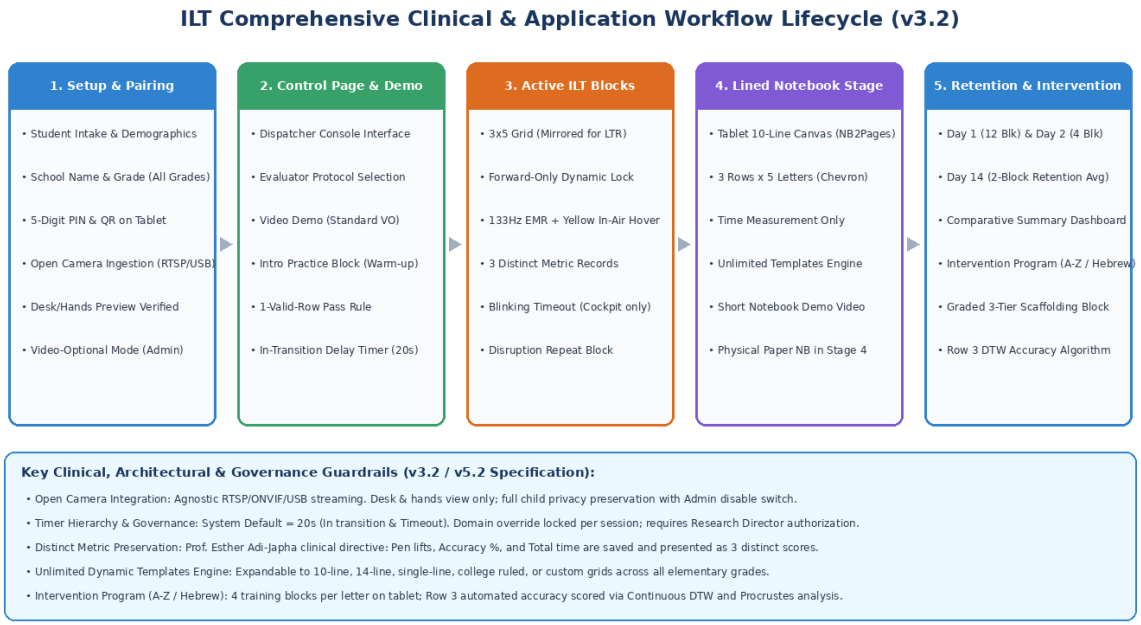


Figure 1: Comprehensive Clinical and System Lifecycle Workflow

3.1 Research Analyst Portal: Kinematic Session Replay & Cohort Analytics

- **Analyst Role & Data Access Protocol:** The diagnostic analyst is not present during testing sessions, but possesses role-based access to the centralized research database. All data is de-identified (protected Participant ID without PII).
- **Individual Kinematic Stroke-by-Stroke Session Replay:** Full kinematic playback: timeline scrubber, speed controls (0.25x, 0.5x, 1x, 2x, 4x, frame step), 133Hz canvas stroke replayer with dynamic pressure heatmaps, yellow in-air hover paths, velocity/jerk profiles, and synchronized desk video playback from MinIO (when recorded).
- **Multi-Dimensional Cohort Slicing & Analytics Dashboard:** Native integration with open-source analytics platforms (Apache Superset, Metabase, DuckDB, Jupyter): multi-dimensional cross-filtering by Domain, School, Grade, Low-SES status, Dominant hand, Language, Session stage, and Timer factoring ratio. Learning curve trajectories, box plots, and one-click Parquet/CSV export.

4. User Interface Specifications & Analyst Wireframes

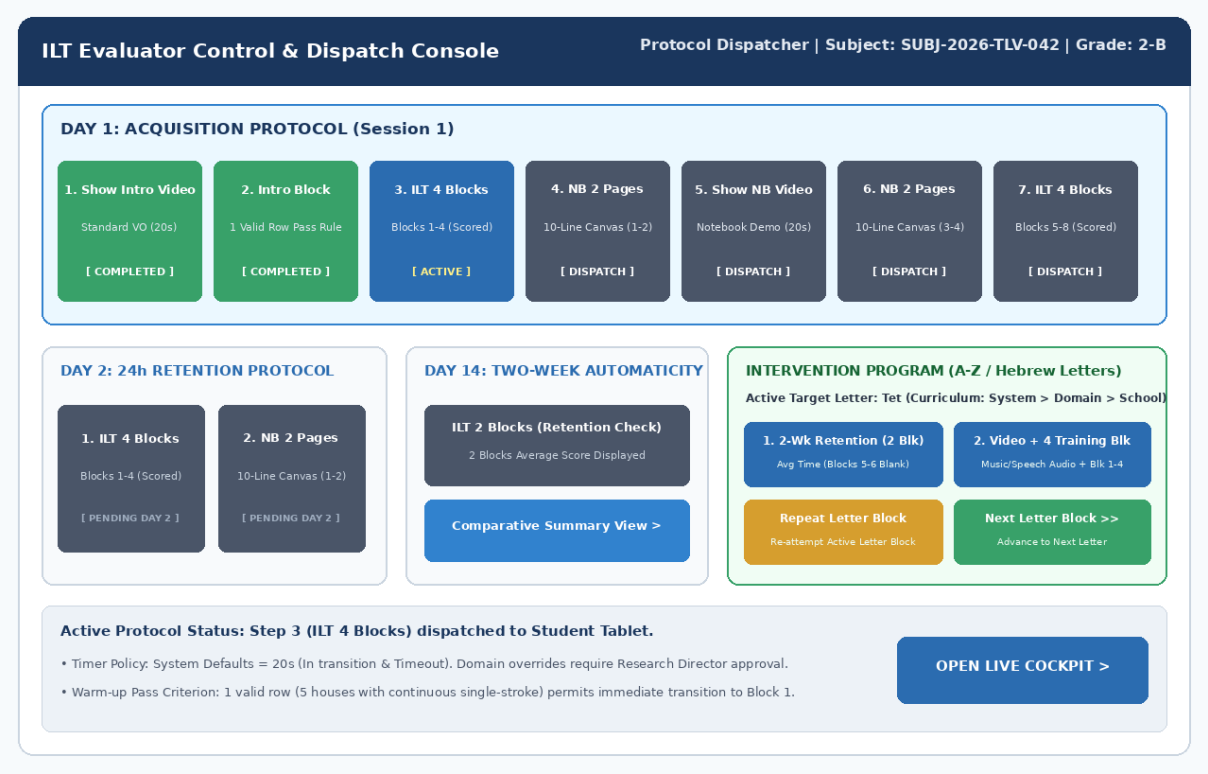


Figure 2: Evaluator Control & Protocol Dispatcher Console

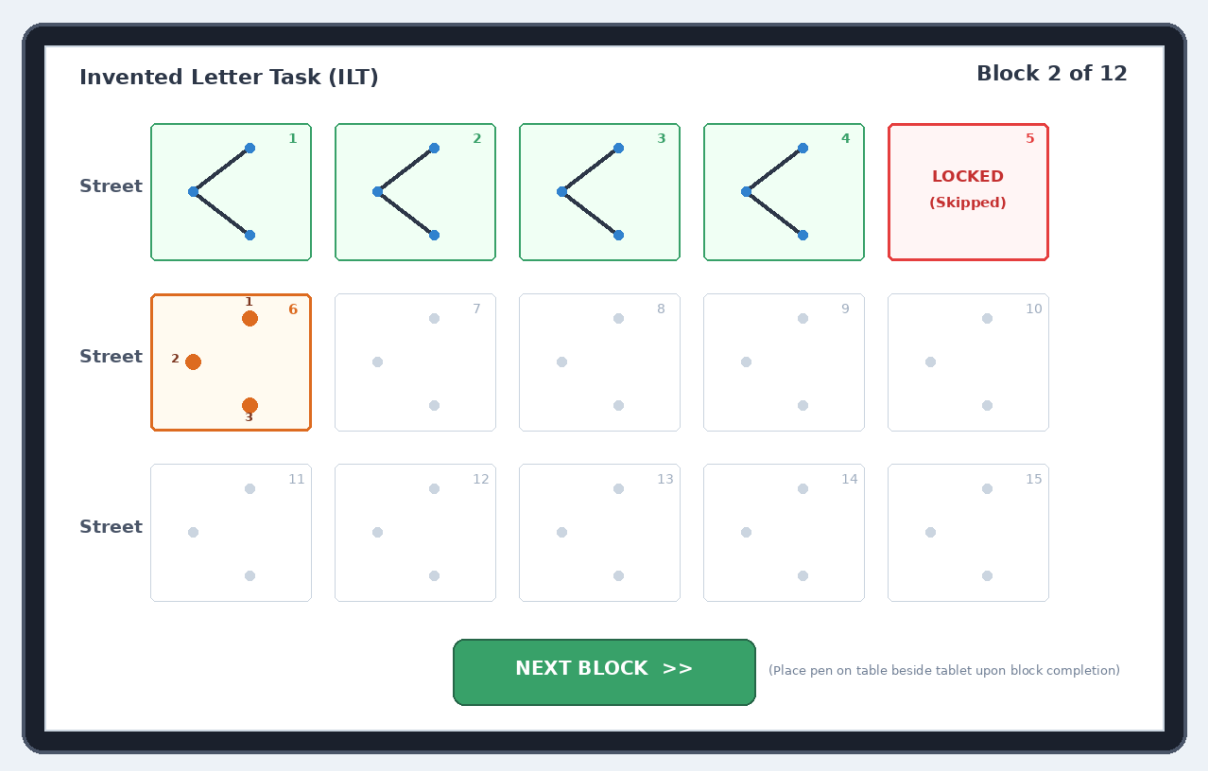


Figure 3: Student Tablet Canvas with Forward-Only Dynamic Lock

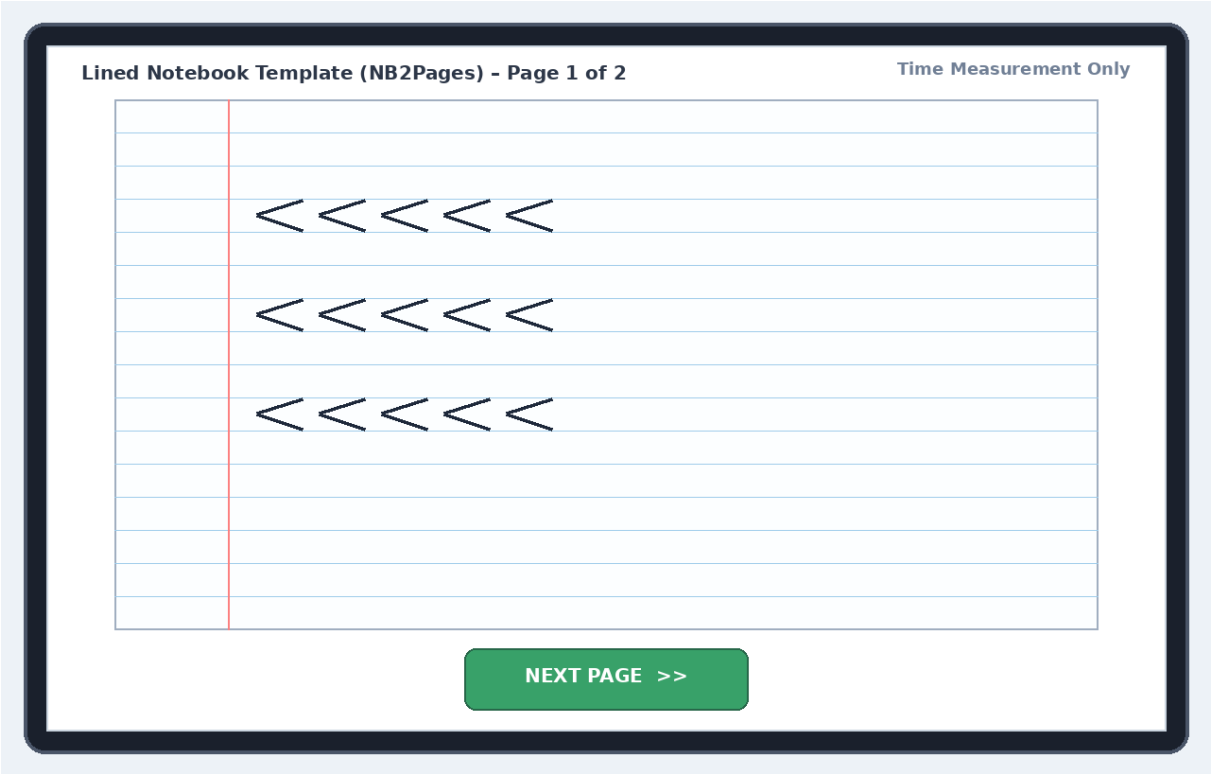


Figure 4: Lined Notebook Canvas Aligned to Original Study Ruling



Figure 5: Proctor Cockpit (80% Live Mirror / 20% Telemetry)

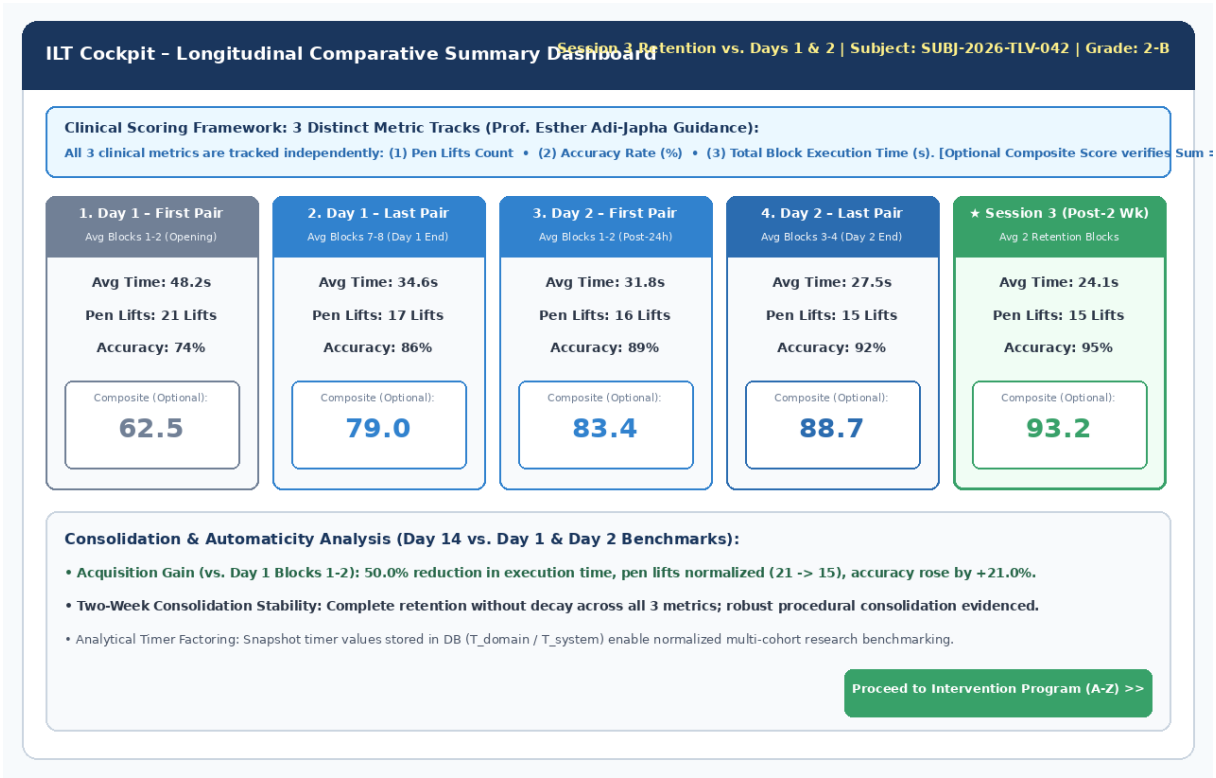


Figure 6: Longitudinal Comparative Summary Dashboard (3 Distinct Metrics)

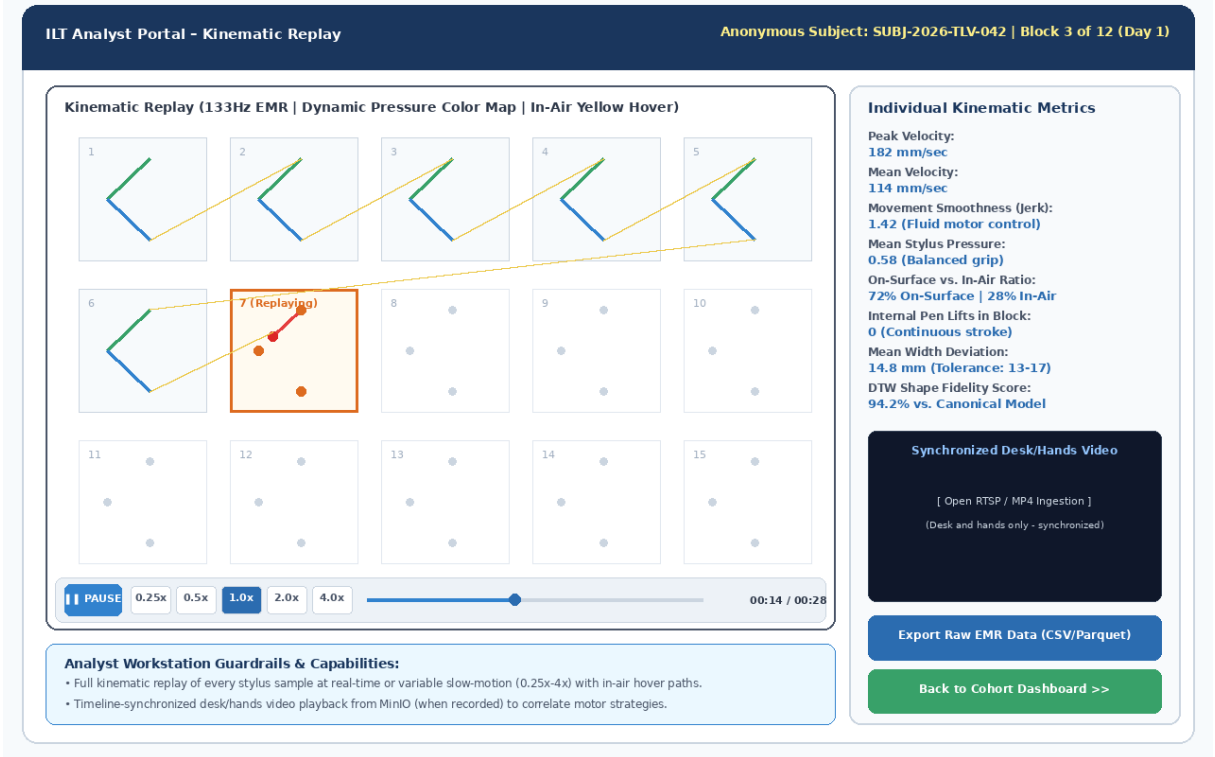


Figure 7: Analyst Portal – Kinematic Session Replay Engine Synced with Desk Video

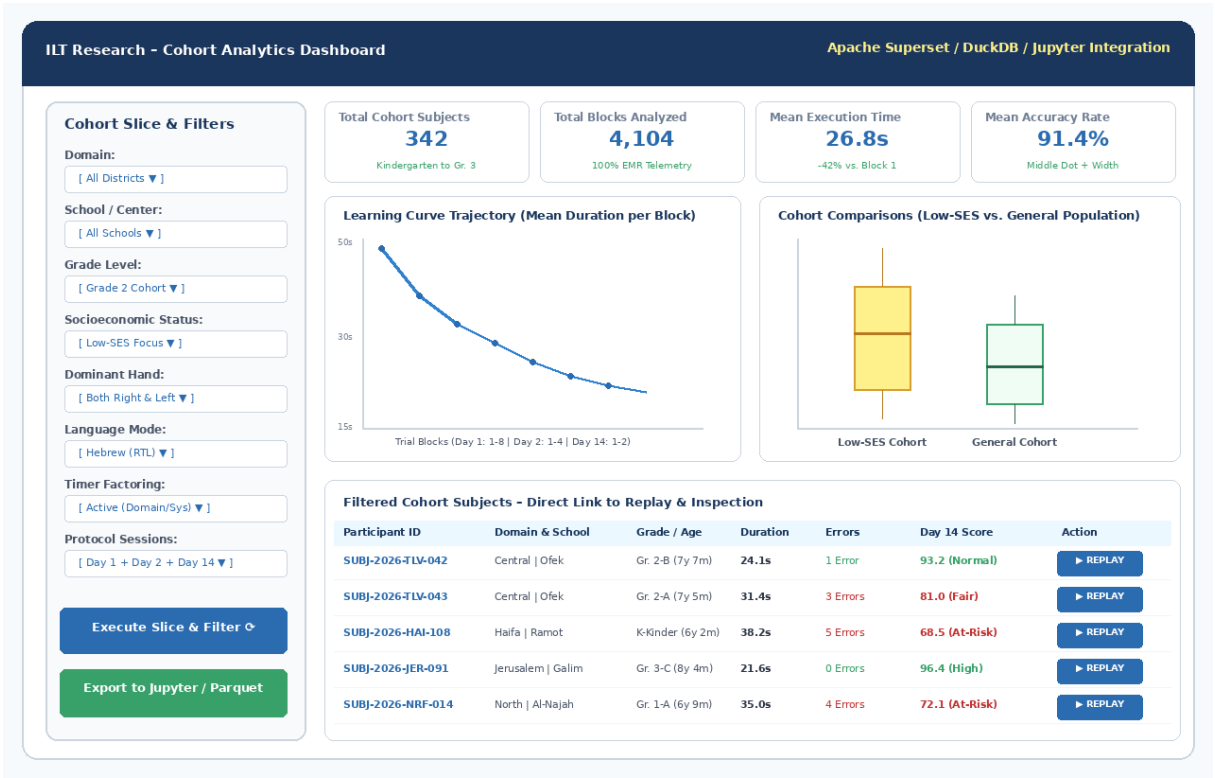


Figure 8: Open-Source Cohort Slice-and-Dice Analytics Dashboard

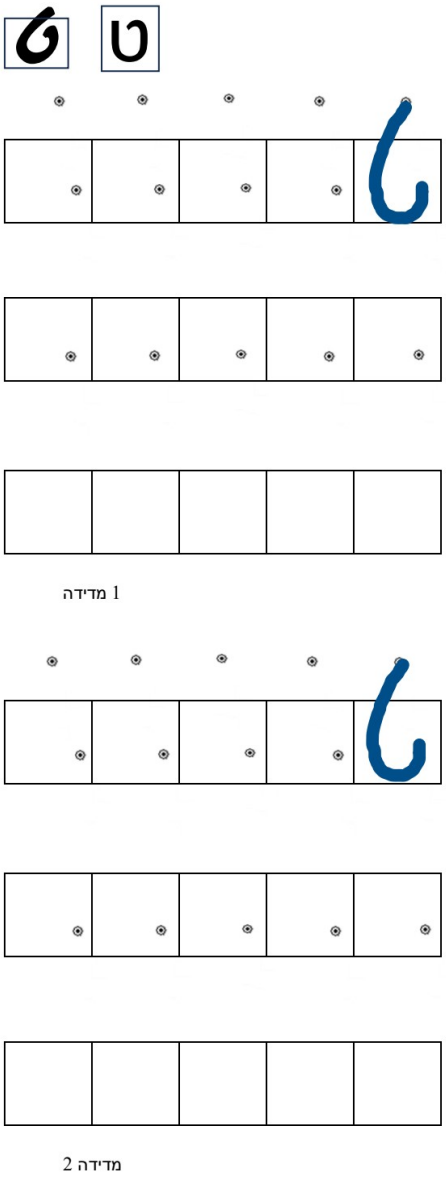


Figure 9: Sample Graded Intervention Training Sheet (Hebrew Letter Tet)

5. Kinematic Error Models & Freehand Row 3 Accuracy

- **cDTW & Procrustes Algorithm for Freehand Row 3:** Resamples stroke to 100 points, computes shape similarity vs. canonical model, validates topology and baseline drift:

```
import numpy as np
from scipy.spatial import procrustes
from fastdtw import fastdtw

def evaluate_freehand_row3(stroke_points: np.ndarray, canonical_points: np.ndarray,
canonical_aspect_ratio: float) -> dict:
    if len(stroke_points) < 10:
        return {"composite_accuracy": 0.0, "diagnostic_tag": "EMPTY_OR_NOISE", "valid": False}

    diffs = np.diff(stroke_points, axis=0)
    dists = np.hypot(diffs[:, 0], diffs[:, 1])
```

```
    cum_dist = np.insert(np.cumsum(dists), 0, 0)
    total_len = cum_dist[-1]
    even_spaces = np.linspace(0, total_len, 100)
    resampled_stroke = np.column_stack((
        np.interp(even_spaces, cum_dist, stroke_points[:, 0]),
        np.interp(even_spaces, cum_dist, stroke_points[:, 1])
    ))

    norm_stroke = (resampled_stroke - np.mean(resampled_stroke, axis=0)) /
(np.std(resampled_stroke) + 1e-6)
    norm_canonical = (canonical_points - np.mean(canonical_points, axis=0)) /
(np.std(canonical_points) + 1e-6)

    _, _, disparity = procrustes(norm_canonical, norm_stroke)
    procrustes_score = max(0.0, min(100.0, 100.0 * (1.0 - disparity * 2.2)))

    dtw_dist, _ = fastdtw(norm_stroke, norm_canonical, dist=2)
    dtw_score = max(0.0, min(100.0, 100.0 * (1.0 - (dtw_dist / 100.0))))

    w, h = np.ptp(stroke_points[:, 0]), np.ptp(stroke_points[:, 1])
    actual_ratio = w / (h + 1e-6)
    ratio_dev_pct = abs(actual_ratio - canonical_aspect_ratio) / canonical_aspect_ratio * 100.0

    composite_acc = 0.55 * procrustes_score + 0.30 * dtw_score + 0.15 * max(0.0, 100.0 -
ratio_dev_pct)
    tag = "ACCURATE" if composite_acc >= 80.0 else ("PROPORTION_DISTORTION" if ratio_dev_pct > 25.0
else "SHAPE_DEFORMATION")

    return {
        "composite_accuracy": round(composite_acc, 2),
        "procrustes_score": round(procrustes_score, 2),
        "dtw_distance": round(dtw_dist, 2),
        "aspect_ratio_deviation_pct": round(ratio_dev_pct, 2),
        "diagnostic_tag": tag,
        "valid": True
    }
```